SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	K.Varshika
Student ID	JF/ICT/24/10
Implementation Status	Completed
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need to export student information for administrative reporting, record-keeping, and external data processing. Previously, student details had to be copied manually from the administrator dashboard when information was required in another application.

CR-015 was introduced to provide a convenient CSV export facility that allows administrators to download student records in a structured format. The exported file can be opened using spreadsheet applications such as Microsoft Excel.

1.2 Existing Behaviour

Before implementation:

- The administrator dashboard loaded student records using `GET /api/students/`.
- Student records were displayed in a table.
- Administrators could search for student records using the existing search functionality.
- Existing Add, Edit, and Delete operations were available.
- No dedicated CSV export facility was available.
- Administrators had to copy student details manually for external reporting.
- Student information was available through the existing student management interface.

1.3 Requested Change

A CSV export facility was required in the administrator dashboard.

The export must:

- Provide an Export CSV button in the administrator dashboard.
- Export student records into a CSV file.
- Include Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- Generate a downloadable file named `students.csv`.
- Support exporting multiple student records.
- Correctly handle commas and quotation marks in field values.
- Display a clear message when no student records are available.
- Preserve existing search, Add, Edit, and Delete functionality.
- Maintain existing authentication and administrator authorization requirements.
- Avoid modifying student records in the database during export.

2. Implementation

2.1 Implementation Summary

The CR-015 requirements were implemented in the administrator dashboard.

An Export CSV button was added to the existing student management interface. When the administrator clicks the button, the application converts the available student records into CSV format and triggers a browser download.

The export function generates a header row containing the six required student fields and then creates a separate row for each available student record. CSV field escaping is used to handle quotation marks and commas correctly.

The generated CSV content is converted into a downloadable file using the browser's Blob functionality. The file is downloaded with the name `students.csv`.

The implementation also includes empty-data handling and styling for the new export button. Existing search, Add, Edit, and Delete functionality remains available.

2.2 Affected Files

File	Change	Purpose
`frontend/src/pages/DashboardPage.jsx`	Modified	Implemented CSV generation, the Export CSV button, file download functionality, and empty-data handling. Preserved existing dashboard functionality.
`frontend/src/styles/dashboard.css`	Modified	Added styling for the Export CSV button, hover effects, focus states, disabled state, and responsive layout.

Additional files	None	No additional files are required for the described frontend implementation.

2.3 Technical Changes

CSV Generation

A CSV generation function was implemented to convert the available student records into comma-separated values.

The exported file contains the following columns:

- Student ID
- Registration Number
- First Name
- Last Name
- Email
- Phone Number

The first row contains the column headings. Each subsequent row contains the corresponding values for one student.

CSV Field Escaping

CSV values are processed to preserve the correct structure of the exported file.

The implementation:

- Encloses field values in quotation marks.
- Escapes quotation marks contained within field values.
- Preserves commas within student information.
- Separates records using new lines.
- Uses UTF-8 encoding with a byte-order mark for improved spreadsheet compatibility.

These operations help prevent values containing commas or quotation marks from being incorrectly separated into multiple columns.

File Download

The generated CSV content is converted into a browser Blob with the appropriate CSV content type.

A temporary object URL is created, and the browser download process is triggered using the filename:

`students.csv`

The temporary object URL is released after the download has been initiated.

Empty-Data Handling

Before generating the CSV file, the application checks whether student records are available.

If the student list is empty, a message informs the administrator that there are no student records to export. This prevents an empty student report from being generated unnecessarily.

User Interface Enhancement

An Export CSV button was added alongside the existing student management controls.

CSS styling was introduced to maintain consistency with the existing dashboard. Hover, focus, and disabled states improve usability, while responsive styling supports smaller screens.

Existing CRUD Preservation

The export feature is intended to be read-only.

Existing Edit and Delete operations continue to use the unique student identifier:

`student.student_id`

Therefore, exporting student information does not change student identities or interfere with existing CRUD operations.

Authentication and Authorization

The export button is provided within the administrator dashboard, which follows the existing authentication flow.

The backend endpoint `GET /api/students/` must enforce administrator authorization to ensure that unauthorized users cannot retrieve protected student records. Frontend button visibility alone does not provide backend security.

2.4 Implementation Logic

1. The administrator signs in to the Student Management System.
2. The administrator dashboard loads student records using the existing `GET /api/students/` request.
3. The application stores the available student records in React state.
4. The administrator clicks the Export CSV button.
5. The application checks whether student records are available.

6. If no records are available, a clear message is displayed.
7. If records are available, the application creates the CSV header row.
8. Each student record is converted into a CSV row containing the six required fields.
9. Field values are escaped to handle commas and quotation marks correctly.
10. The CSV content is converted into a downloadable Blob.
11. The browser downloads the generated file as `students.csv`.
12. The administrator can open the downloaded file using a compatible spreadsheet application.
13. Existing search, Add, Edit, and Delete functionality continues to operate independently of the export process.

2.5 Error and Data Handling

The implementation handles empty student lists and CSV field values containing special characters.

The following conditions are considered:

- An empty student list displays a clear message.
- Multiple records are processed as separate CSV rows.
- Commas and quotation marks are escaped within field values.
- UTF-8 encoding improves compatibility with non-English text.
- Missing or null values should be represented safely in the generated CSV.
- Student-loading errors continue to be handled by the existing API error-handling logic.
- Exporting records does not create, update, or delete database records.

The export uses the student records available in the frontend. If the backend returns a paginated dataset, the export contains only the records loaded by the frontend unless all records are retrieved separately.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual testing of the administrator dashboard and the generated CSV file.

Testing Types:

- Functional testing
- Positive testing
- CSV format validation
- Data integrity testing
- Empty-data testing
- UI testing
- Regression testing

- Authentication and authorization testing
- CRUD verification

Environment: Development

Browser: Google Chrome

Test Role: Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Verify the Export CSV button	The Export CSV button is displayed in the administrator dashboard.	Export CSV button displayed correctly.	Pass
TC-02	Export a single student record	A CSV file containing the student's information is generated.	Student record exported successfully.	Pass
TC-03	Export multiple student records	All available loaded student records are exported.	Multiple records exported successfully.	Pass
TC-04	Verify CSV column headings	The file contains all six required column headings.	All required headings displayed correctly.	Pass
TC-05	Verify the downloaded filename	The downloaded file is named `students.csv`.	File downloaded with the correct filename.	Pass
TC-06	Export records containing commas or quotation marks	Field values remain correctly formatted in the CSV file.	Special characters handled correctly.	Pass
TC-07	Open the CSV in a spreadsheet application	Student information appears in the appropriate columns.	CSV opened and columns displayed correctly.	Pass
TC-08	Export when the student list is empty	A clear no-records message is displayed.	No-records message displayed correctly.	Pass
TC-09	Verify existing search functionality	Search and Clear Search continue to work.	Existing search functionality worked correctly.	Pass
TC-10	Verify Add, Edit, and Delete operations	Existing CRUD operations continue to work correctly.	CRUD operations worked correctly.	Pass
TC-11	Verify authentication and authorization	Protected student data is available only to authorized users.	Existing authentication and authorization verified.	Pass
TC-12	Verify database integrity after export	Exporting does not modify student database records.	Student records remained unchanged after export.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	An Export CSV button is available in the administrator dashboard.	Dashboard UI testing	Met
AC-002	Student records can be exported to a CSV file.	Functional export testing	Met
AC-003	The CSV contains all six required column headings.	CSV header verification	Met
AC-004	Student field values are exported into the correct columns.	Data comparison testing	Met
AC-005	The downloaded file is named `students.csv`.	Download verification	Met
AC-006	Multiple student records can be exported.	Multiple-record export testing	Met
AC-007	Empty student data is handled with a clear message.	Empty-data testing	Met
AC-008	CSV special characters are handled correctly.	CSV format testing	Met
AC-009	Existing search and CRUD functionality continues to work.	Regression testing	Met
AC-010	Exporting records does not modify database records.	Database integrity testing	Met
AC-011	Existing authentication and authorization requirements are preserved.	Authentication and authorization testing	Met

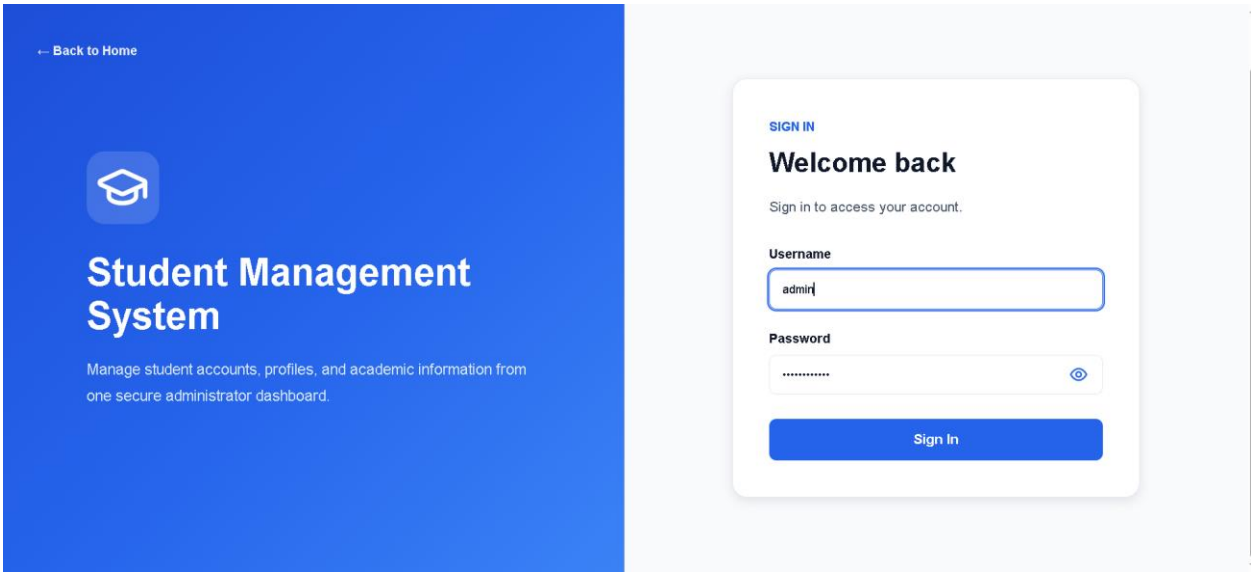
5. Evidence

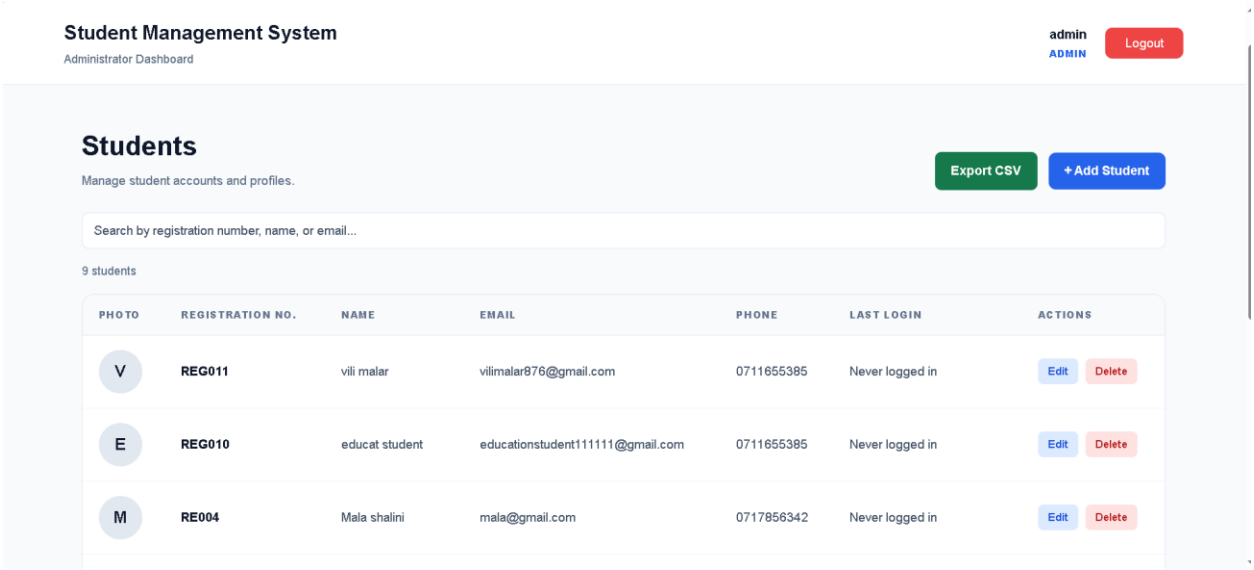
A maximum of two relevant screenshots should be attached.

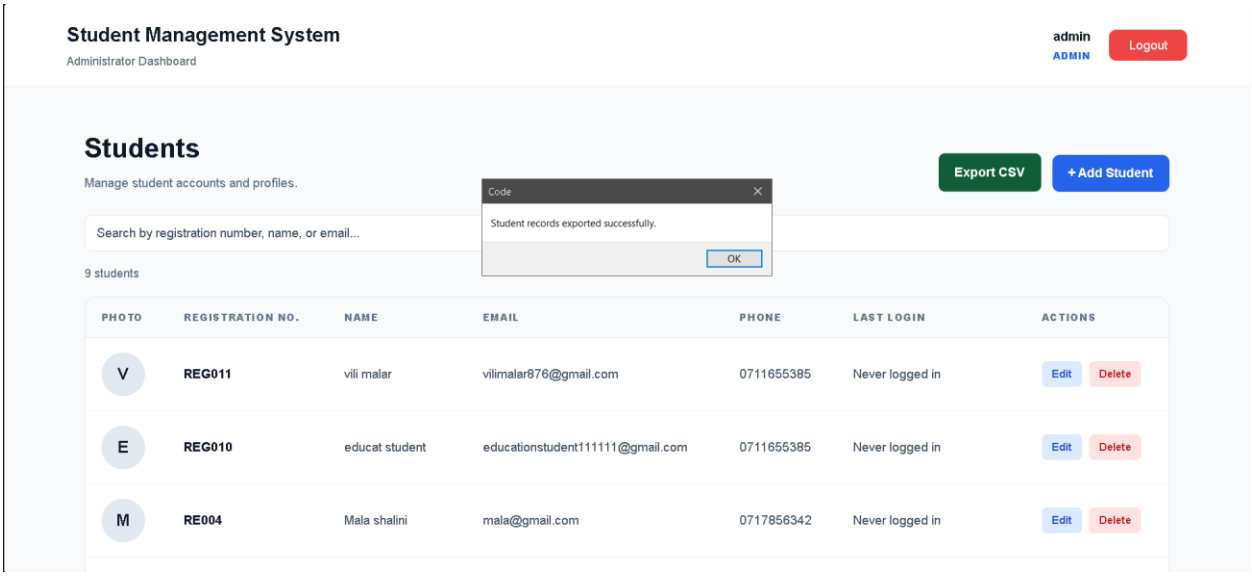
Evidence 01 – Export CSV Functionality

Related: AC-001, AC-002, AC-003, AC-005, AC-006

Figure 1: Administrator dashboard showing the Export CSV button and the available student records before exporting.



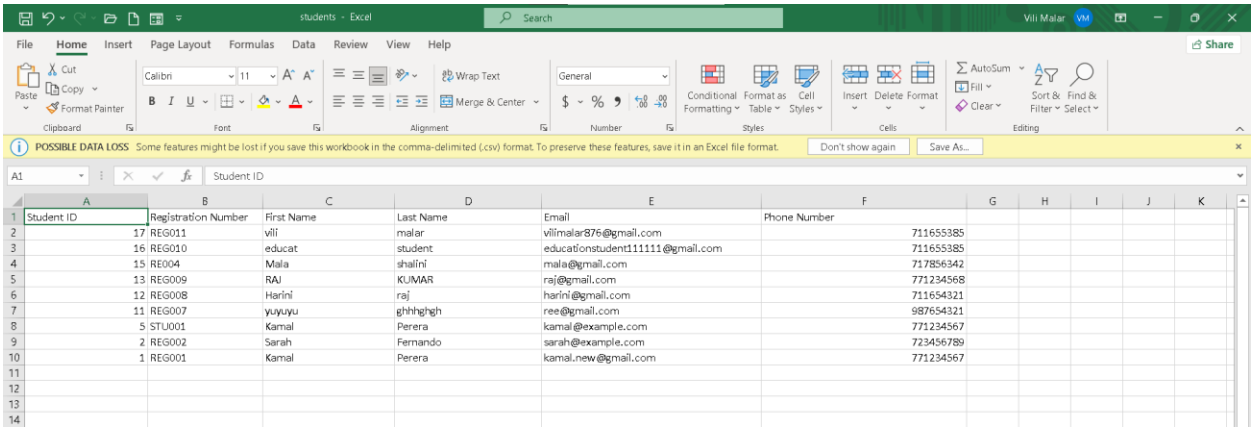




Evidence 02 – Downloaded CSV File

Related: AC-003, AC-004, AC-005, AC-008

Figure 2: The downloaded `students.csv` file opened in a spreadsheet application, showing the six required column headings and the exported student records.



6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Authentication and Authorization	Verified

Final Status	Accepted

6.1 Final Summary

CR-015 was implemented as a student records export enhancement for the administrator dashboard of the Student Management System.

The implementation provides an Export CSV button that allows administrators to download available student records in a CSV file named `students.csv`. The exported file contains Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.

The CSV generation process handles field values containing commas and quotation marks and uses UTF-8 encoding for improved compatibility with spreadsheet applications. Empty-data handling was also included to inform administrators when no student records are available for export.

The feature uses the student records already loaded into the frontend and does not require a database schema modification. Existing search, Add, Edit, and Delete functionality is preserved. Exporting is designed as a read-only operation and does not modify student records.

The implementation was verified against the defined test cases and acceptance criteria. All 12 test cases passed, giving a pass rate of 100%. All acceptance criteria were marked as met, and the overall implementation status was recorded as Completed.

6.2 Issues / Limitations

No significant issues identified during testing.

The export contains the student records available in the frontend. If the API uses pagination, exporting every database record may require additional backend support.

The backend must continue to enforce administrator authorization to protect student information.

Student Name: K.Varshika

Student ID: JF/ICT/24/10

Date: 15 October 2026

Signature: K.Varshika

~ END OF REPORT ~